

Retail Project

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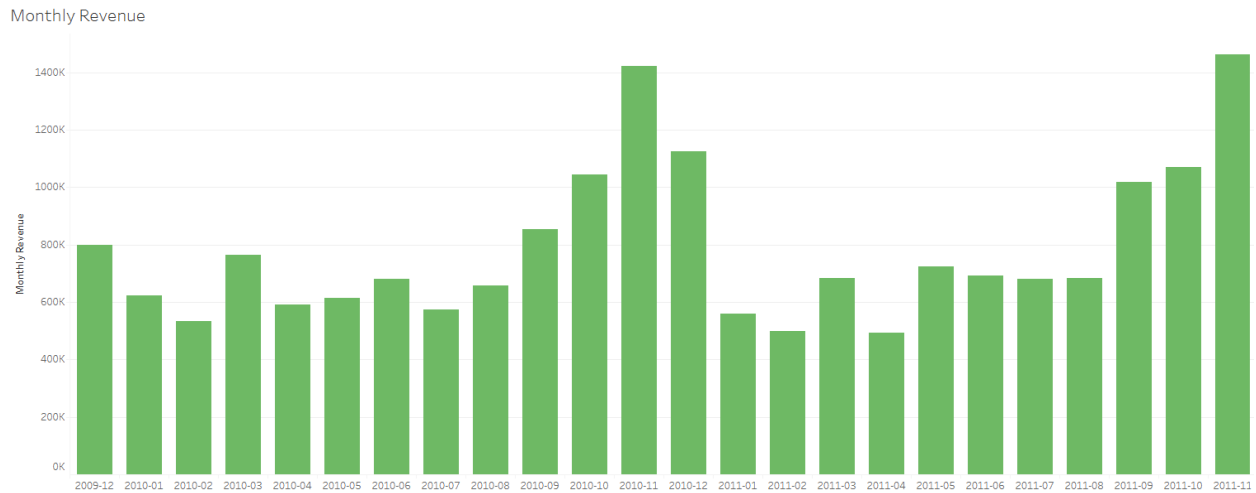
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Project Overview:

This is a transactional dataset that contains all the transactions occurring between 12/01/2009 and 12/09/2011 for a UK-based and registered non-store online retailer. The company mainly sells unique all-occasion gifts. Many customers of the company are wholesalers. For this project, I used MySQL, Python, and Tableau.

Question 1: Is revenue growing, flat, or declining over the dataset's time period?



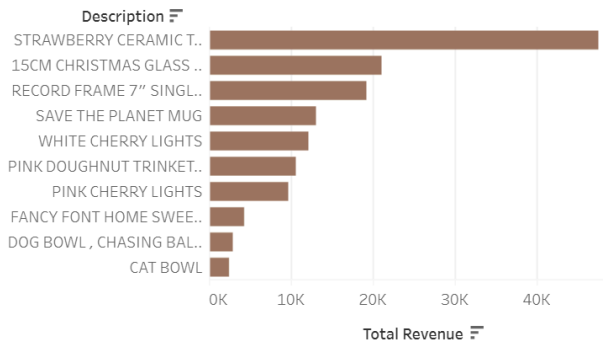
Findings:

Looking at the chart listed above, there is no specific trend of revenue growing, decreasing, or being flat throughout the dataset's time period. It is quite stable throughout the months, with large spikes, most notably in November (both 2010-11 & 2011-11). Revenue starts to increase steadily in September and decreases after the November peak, continuing into December. Since the company sells unique gifts, peak sales are expected around the holiday season. Outside of these peaks, it stays steady, with a noticeable decrease after the holiday due to the season being over.

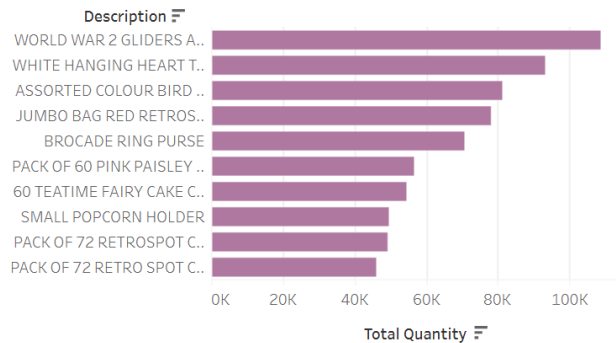
Suggestion: If the company knows that sales peak around this time, specifically the holiday season, there should be incentives to purchase more for wholesale companies, such as discounts. Although, as noted in question 3, it may be difficult to distinguish the difference between wholesale companies and frequent buyers. Further analysis will be required for this. The company should also increase its purchase of inventory items to be able to keep up with the volume of orders.

Question 2: What are the top 10 best-selling products by quantity vs. by revenue?

Top 10 Products by Revenue



Top 10 Products by Quantity



Findings:

There is no overlap between the top 10 products by quantity sold and the top 10 products by revenue generated; these are two completely distinct sets of products. The top sellers by quantity are dominated by bulk-pack items (e.g., packs of 60 or 72 units), suggesting these are low-cost, high-volume items likely purchased in large numbers per order. In contrast, the top products by revenue are largely decorative and gift items, things like lights and mugs, indicating these carry a higher per-unit price and generate revenue without necessarily selling in huge quantities.

Suggestions:

Since bulk-pack items drive volume but not necessarily profit, the company should evaluate whether these products are functioning as effectively as they should be for their customer base or whether they're eating into margin without a clear strategic purpose. Meanwhile, higher-revenue items like decorations and gift pieces may benefit from increased marketing visibility or better placement, since they're valued by customers at a higher price point without needing bulk volume to perform well.

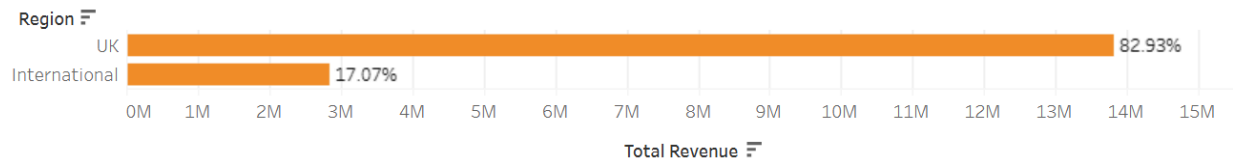
Question 3: Since "many customers are wholesalers," can you distinguish wholesale vs. individual buyers by looking at order size (e.g., customers who consistently buy in bulk quantities)?

Findings: To distinguish between wholesale and individual buyers, I counted the average quantity purchased per order and the total quantity purchased per customer overall. In both cases, a single customer was an extreme outlier, averaging roughly 12,000 units per order and holding a total purchase volume roughly 15,000 units higher than the next-highest customer. Beyond this one outlier, the remaining customer base showed a continuous, gradual distribution in purchase quantity, with no clear secondary tier or natural break point that would indicate a broader "wholesale" segment.

Suggestion: Rather than attempting to segment customers into "wholesale" vs. "individual" based on order size alone, the company would benefit from perhaps tracking customer type at the point of account creation, since purchase-quantity patterns alone don't separate the two groups in this dataset. Although the one high-volume outlier customer identified here may warrant direct attention, given their significant purchase volume relative to the rest of the customer base.

Question 4: What % of revenue comes from the UK vs. international customers?

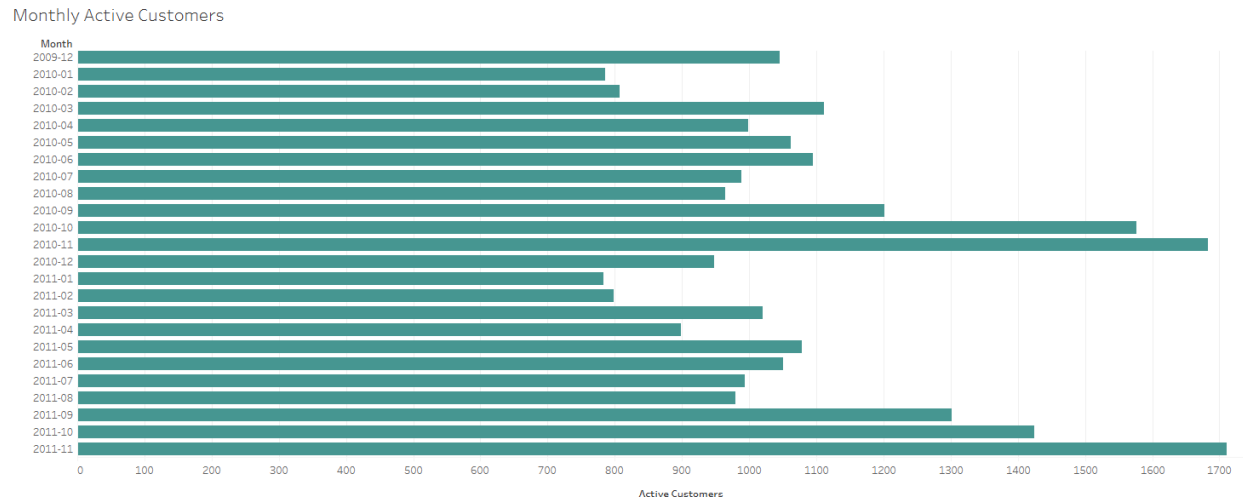
% of revenue - UK vs. International



Findings: The business is heavily concentrated in the UK market, with approximately 83% of total revenue coming from UK customers and only 17% from international customers. While the company does sell internationally, this split shows the business is still fundamentally UK-driven rather than having a large international customer base.

Suggestion: Since revenue is still mainly concentrated in the UK, the company should either focus on bettering their marketing strategy and treat international sales as a genuine growth opportunity and worthwhile investment, or recognize that international demand may be naturally limited by shipping costs and logistics for a gift retailer, in which case resources may be better spent focusing on retention within the UK customer base. The company should also look at which specific non-UK countries contribute the most to that 17%, and if one or two countries stand out, that could indicate a more targeted, lower-risk expansion opportunity rather than broad international investment.

Question 5: What does the customer retention pattern look like month over month? Are customers from early in the dataset still buying by the end?



Findings: Active customer counts follow the same seasonal pattern seen in revenue. Customer activity rises steadily starting in September, peaks sharply in November (both 2010 and 2011), and drops off afterward. This confirms that the November revenue spike isn't just existing customers spending more; it's that more customers are engaging with the business during this period. Outside the seasonal peak, monthly active customer counts remain relatively stable, without a clear long-term growth or decline trend across the full dataset.

It's worth noting this measures active customers per month, not true customer retention. This metric shows overall engagement volume, but a deeper retention analysis would be needed to understand how well the business retains specific customers over time.

Suggestion: Since customer volume, not just spend, spikes seasonally, the company should ensure customer service and support capacity scales alongside inventory during peak months, since support during the highest-traffic period could damage the customer experience precisely when the most new and returning customers are engaging with the brand. As a next step, retention analysis would provide a clearer picture of long-term customer loyalty beyond month-to-month.